Exploratory Data Analysis (EDA)

Rx Drug Marketing Mix Modeling (MMIX)

Business Context

Pharmaceutical firms compete within therapeutic classes, where multiple brands seek to influence prescriptions through two primary channels:

- **Detailing**: physician-targeted promotion
- **DTCA**: direct-to-consumer advertising

The commercial mechanism is:

> Marketing investment → influences physicians and patients → drives prescriptions → generates sales

Business Objective

The goal of this analysis is to understand how marketing investments translate into sales outcomes across brands and therapeutic classes, and to identify patterns that should guide model design and budget decisions.

Core Business Questions

1. Which brands dominate portfolio revenue, and how concentrated is the portfolio?
2. How has portfolio performance evolved over time?
3. How are detailing and Direct-to-Consumer Ads (DTCA) budgets allocated across brands and classes?
4. Which channel appears more efficient before formal modeling?
5. Is there evidence of diminishing returns to marketing spend?
6. Do classes and brands differ in baseline sales and marketing sensitivity?
7. Are there structurally different marketing regimes, especially around zero Direct-to-Consumer Ads (DTCA)?
8. What does the panel structure imply for modeling and validation?

Scope of Analysis

This review encompasses an 11-year historical dataset, capturing the following parameters:

- Observation Period: 2013–2024
- Therapeutic Classes: PPI (Proton Pump Inhibitors), SSRI (Antidepressants), Statin (Cholesterol)
- Portfolio Size: 22 distinct brands
- Key Metrics Evaluated:
  - Top-line Sales in Billions of Dollars
  - Physician Detailing Spend in Millions of Dollars
  - Direct-to-Consumer Advertising (DTCA) Spend ($ Millions)

In [240…

```
print("\n[EDA] Importing libraries...")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
pd.set_option("display.max_columns", 100)
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")

print("[EDA] Libraries loaded successfully.")
```

```
[EDA] Importing libraries...
[EDA] Libraries loaded successfully.
```

In [241…

```
print("\n[EDA] Loading dataset and standardizing column names...")

df = pd.read_excel("data/MMM_Drug_Data.xlsx")
```

```python
df = df.rename(columns={
    "Class": "class",
    "Agent": "agent",
    "Year": "year",
    "Detailing": "detailing",
    "Direct-to-Consumer Advertising (DTCA)": "direct_to_consumer_ad",
    "Actual Sales": "actual_sales"
})

print("[EDA] Dataset loaded successfully.")
display(df.head())
```

```
[EDA] Loading dataset and standardizing column names...
[EDA] Dataset loaded successfully.
```

	class	agent	year	detailing	direct_to_consumer_ad	actual_sales
0	PPI	Prevacid	2013	43.8805	0.0000	2.9083
1	PPI	Prilosec	2013	41.6480	0.0000	14.7235
2	SSRI	Luvox	2013	22.3667	0.0000	0.9120
3	SSRI	Paxil	2013	57.4608	0.0000	11.2677
4	SSRI	Prozac	2013	84.5333	0.0000	20.2600

In [242…

```python
print("\n[EDA] Running initial data checks...")

print("Shape:", df.shape)

print("\nColumn types:")
display(df.dtypes.to_frame("dtype"))

print("\nMissing values:")
display(df.isna().sum().to_frame("missing_count"))

print("\nYears covered:", df["year"].min(), "to", df["year"].max())
print("Number of classes:", df["class"].nunique())
print("Number of agents:", df["agent"].nunique())

print("\n[Business Insight]")
print("The dataset covers 12 years (2013-2024), 22 agents (brands), and 3 therapeutic classes")
print("PPI (Proton Pump Inhibitors), SSRI (Antidepressants), Statin (Cholesterol) classes have zero mis
print("This gives sufficient depth to evaluate marketing effectiveness at both the agent (brand) and cl
```

```
[EDA] Running initial data checks...
Shape: (207, 6)

Column types:
```

	dtype
class	str
agent	str
year	int64
detailing	float64
direct_to_consumer_ad	float64
actual_sales	float64

```
Missing values:
```

	missing_count
class	0
agent	0
year	0
detailing	0
direct_to_consumer_ad	0
actual_sales	0

```
Years covered: 2013 to 2024
Number of classes: 3
Number of agents: 22

[Business Insight]
The dataset covers 12 years (2013–2024), 22 agents (brands), and 3 therapeutic classes
PPI (Proton Pump Inhibitors), SSRI (Antidepressants), Statin (Cholesterol) classes have zero missing val
ues.
This gives sufficient depth to evaluate marketing effectiveness at both the agent (brand) and class leve
l.
```

In [243…

```
print("\n[EDA] Checking agent (brand) coverage across years...")

panel = pd.crosstab(df["agent"], df["year"])

n_agents = df["agent"].nunique()
n_years = df["year"].nunique()
expected_rows = n_agents * n_years
actual_rows = len(df)

print(f"Expected rows in a balanced panel: {expected_rows}")
print(f"Actual rows observed: {actual_rows}")
print(f"Missing agent-year combinations: {expected_rows - actual_rows}")

display(panel)

# Diagnostic only: checking if any agents (brands) appear in later years but not in earlier years.
# Time-based split is intentional: random splits leak future data into training.
# Final modeling split: Train 2013–2021, Validation 2022–2023, Test 2024.
test_agents = set(df[df["year"] >= 2023]["agent"].unique())
train_agents = set(df[df["year"] <= 2020]["agent"].unique())

print("\n[EDA] Checking if any agents (brands) appear in test years but not in training years...")
print(f"Brands in test period (2023–2024) with no data in training period (<=2020): {test_agents - trai
print(f"Brands in training period (<=2020) with no data in test period (2023–2024): {train_agents - tes

print("\n[Business Insight]")
print("Only 207 of 264 possible agent (brand)-year combinations exist. Five agents (brands) entered aft
print("and Pexeva only appears from 2021 (4 records). Historical trends for newer agents (brands) are s
print("and less reliable for forecasting or head-to-head comparisons with agents (brands) that have 12
print("[Modeling Note] Time-based split: Train 2013–2021, Validation 2022–2023, Test 2024. Pexeva requi
```

```
[EDA] Checking agent (brand) coverage across years...
Expected rows in a balanced panel: 264
Actual rows observed: 207
Missing agent-year combinations: 57
```

year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
agent												
Aciphex	0	0	0	1	1	1	1	1	1	1	1	1
Altoprev	0	0	0	0	0	0	1	1	1	1	1	1
Celexa	0	0	1	1	1	1	1	1	1	1	1	1
Crestor	0	0	0	0	0	0	0	1	1	1	1	1
Lescol	1	1	1	1	1	1	1	1	1	1	1	1
Lescol XL	0	0	0	0	1	1	1	1	1	1	1	1
Lexapro	0	0	0	0	0	0	1	1	1	1	1	1
Lipitor	0	1	1	1	1	1	1	1	1	1	1	1
Luvox	1	1	1	1	1	1	1	1	1	1	1	1
Mevacor	1	1	1	1	1	1	1	1	1	1	1	1
Nexium	0	0	0	0	0	1	1	1	1	1	1	1
Paxil	1	1	1	1	1	1	1	1	1	1	1	1
Paxil CR	0	0	0	0	0	0	1	1	1	1	1	1
Pexeva	0	0	0	0	0	0	0	0	1	1	1	1
Pravachol	1	1	1	1	1	1	1	1	1	1	1	1
Prevacid	1	1	1	1	1	1	1	1	1	1	1	1
Prilosec	1	1	1	1	1	1	1	1	1	1	1	1
Protonix	0	0	0	0	1	1	1	1	1	1	1	1
Prozac	1	1	1	1	1	1	1	1	1	1	1	1
Prozac Weekly	0	0	0	0	0	1	1	1	1	1	1	1
Zocor	1	1	1	1	1	1	1	1	1	1	1	1
Zoloft	1	1	1	1	1	1	1	1	1	1	1	1

```
[EDA] Checking if any agents (brands) appear in test years but not in training years...
Brands in test period (2023-2024) with no data in training period (<=2020): {'  Pexeva'}
Brands in training period (<=2020) with no data in test period (2023-2024): None

[Business Insight]
Only 207 of 264 possible agent (brand)-year combinations exist. Five agents (brands) entered after 2016,
and Pexeva only appears from 2021 (4 records). Historical trends for newer agents (brands) are shorter
and less reliable for forecasting or head-to-head comparisons with agents (brands) that have 12 years of
data.
[Modeling Note] Time-based split: Train 2013-2021, Validation 2022-2023, Test 2024. Pexeva requires spec
ial handling.
```

In [244…

```
print("\n[EDA] Generating summary statistics for key commercial variables...")

summary = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].describe()

summary_readable = summary.copy()
summary_readable["actual_sales"] = summary["actual_sales"].apply(lambda x: f"${x:,.2f}B")
summary_readable["detailing"] = summary["detailing"].apply(lambda x: f"${x:,.2f}M")
summary_readable["direct_to_consumer_ad"] = summary["direct_to_consumer_ad"].apply(lambda x: f"${x:,.2f

summary_readable = summary_readable.reset_index().rename(columns={"index": "Metric"})
display(summary_readable)

print("\n[Business Insight]")
print("Sales range from ~$0B to $63B; detailing from $0 to $243M; Direct-to-Consumer Ads (DTCA) from $0
print("The median agent (brand) generates $7.5B with $45M detailing and $0 Direct-to-Consumer Ads (DTCA
print("meaning most agents (brands) do not advertise to consumers at all.")
print("A small number of high-revenue agents (brands) dominate the portfolio averages.")
```

```
[EDA] Generating summary statistics for key commercial variables...
```

	Metric	actual_sales	detailing	direct_to_consumer_ad
0	count	$207.00B	$207.00M	$207.00M
1	mean	$11.67B	$56.03M	$26.32M
2	std	$12.97B	$54.92M	$49.30M
3	min	$0.00B	$0.00M	$0.00M
4	25%	$0.99B	$3.86M	$0.00M
5	50%	$7.53B	$44.97M	$0.00M
6	75%	$19.61B	$85.12M	$39.36M
7	max	$63.22B	$243.33M	$240.35M

```
[Business Insight]
Sales range from ~$0B to $63B; detailing from $0 to $243M; Direct-to-Consumer Ads (DTCA) from $0 to $240
M.
The median agent (brand) generates $7.5B with $45M detailing and $0 Direct-to-Consumer Ads (DTCA),
meaning most agents (brands) do not advertise to consumers at all.
A small number of high-revenue agents (brands) dominate the portfolio averages.
```

1. Revenue Concentration

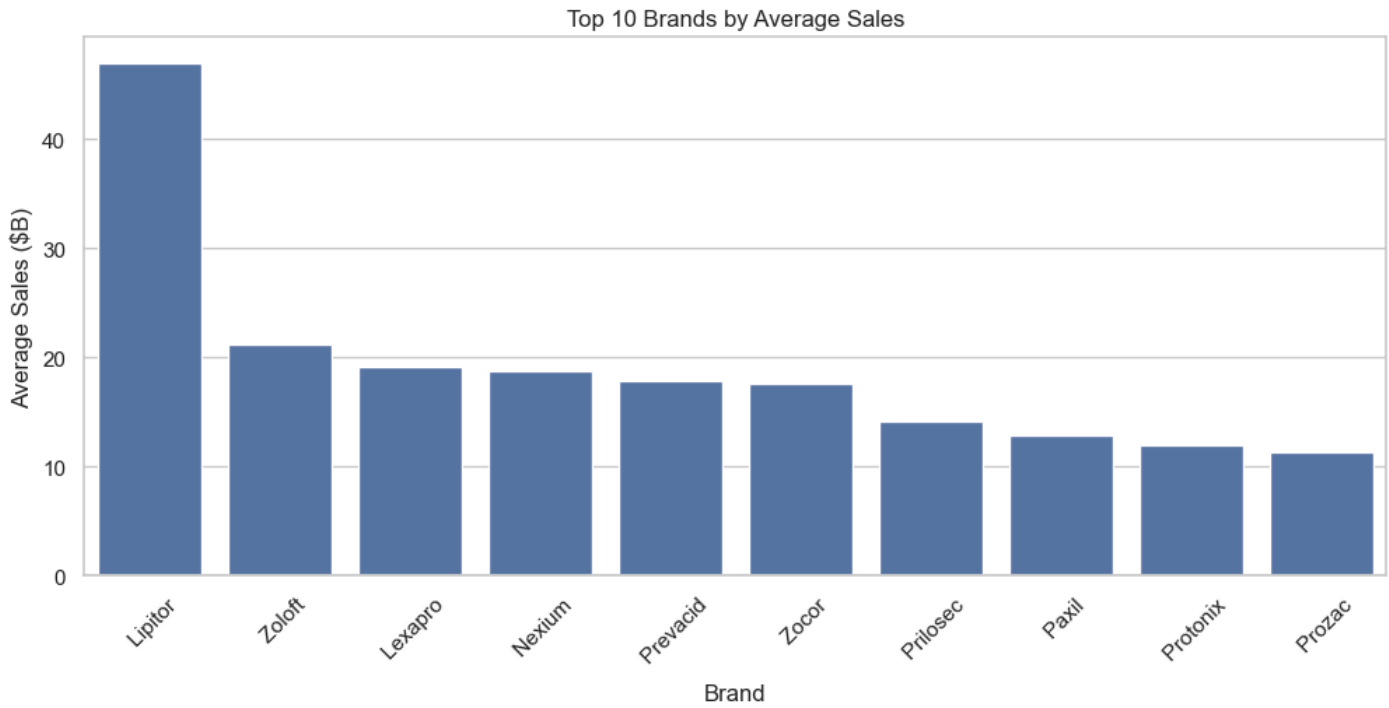
A first strategic question is whether sales are diversified or concentrated in a few blockbuster brands.

In [245…

```
print("\n[EDA] Ranking brands by average sales contribution...")

brand_sales = (
    df.groupby("agent", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

brand_sales["sales_share_pct"] = 100 * brand_sales["actual_sales"] / brand_sales["actual_sales"].sum()
brand_sales["cumulative_share_pct"] = brand_sales["sales_share_pct"].cumsum()

display(brand_sales.head(10))

plt.figure(figsize=(12, 5))
sns.barplot(data=brand_sales.head(10), x="agent", y="actual_sales")
plt.title("Top 10 Brands by Average Sales")
plt.xlabel("Brand")
plt.ylabel("Average Sales ($B)")
plt.xticks(rotation=45)
plt.show()

top_5_share = brand_sales.head(5)["sales_share_pct"].sum()

print("\n[Business Insight]")
print(f"The top 5 agents (brands) (Lipitor, Zoloft, Lexapro, Nexium, Prevacid) account for {top_5_share
print("Lipitor alone contributes ~20%. A 10% decline in Lipitor has roughly the same revenue impact")
print("as losing the bottom 8 agents (brands) entirely.")
print("Budget decisions around these high-revenue agents (brands) carry outsized portfolio risk.")
```

```
[EDA] Ranking brands by average sales contribution...
```

	agent	actual_sales	sales_share_pct	cumulative_share_pct
7	Lipitor	47.0275	19.7929	19.7929
21	Zoloft	21.1441	8.8991	28.6920
6	Lexapro	19.1402	8.0557	36.7477
10	Nexium	18.7587	7.8951	44.6428
15	Prevacid	17.8747	7.5231	52.1659
20	Zocor	17.6050	7.4096	59.5754
16	Prilosec	14.1557	5.9578	65.5333
11	Paxil	12.8184	5.3950	70.9283
17	Protonix	12.0249	5.0610	75.9893
18	Prozac	11.2791	4.7471	80.7364




```
[Business Insight]
The top 5 agents (brands) (Lipitor, Zoloft, Lexapro, Nexium, Prevacid) account for 52.2% of portfolio re
venue.
Lipitor alone contributes ~20%. A 10% decline in Lipitor has roughly the same revenue impact
as losing the bottom 8 agents (brands) entirely.
Budget decisions around these high-revenue agents (brands) carry outsized portfolio risk.
```

2. Growth Trajectory

Before asking which marketing works, we need to understand whether the portfolio is expanding, maturing, or declining.

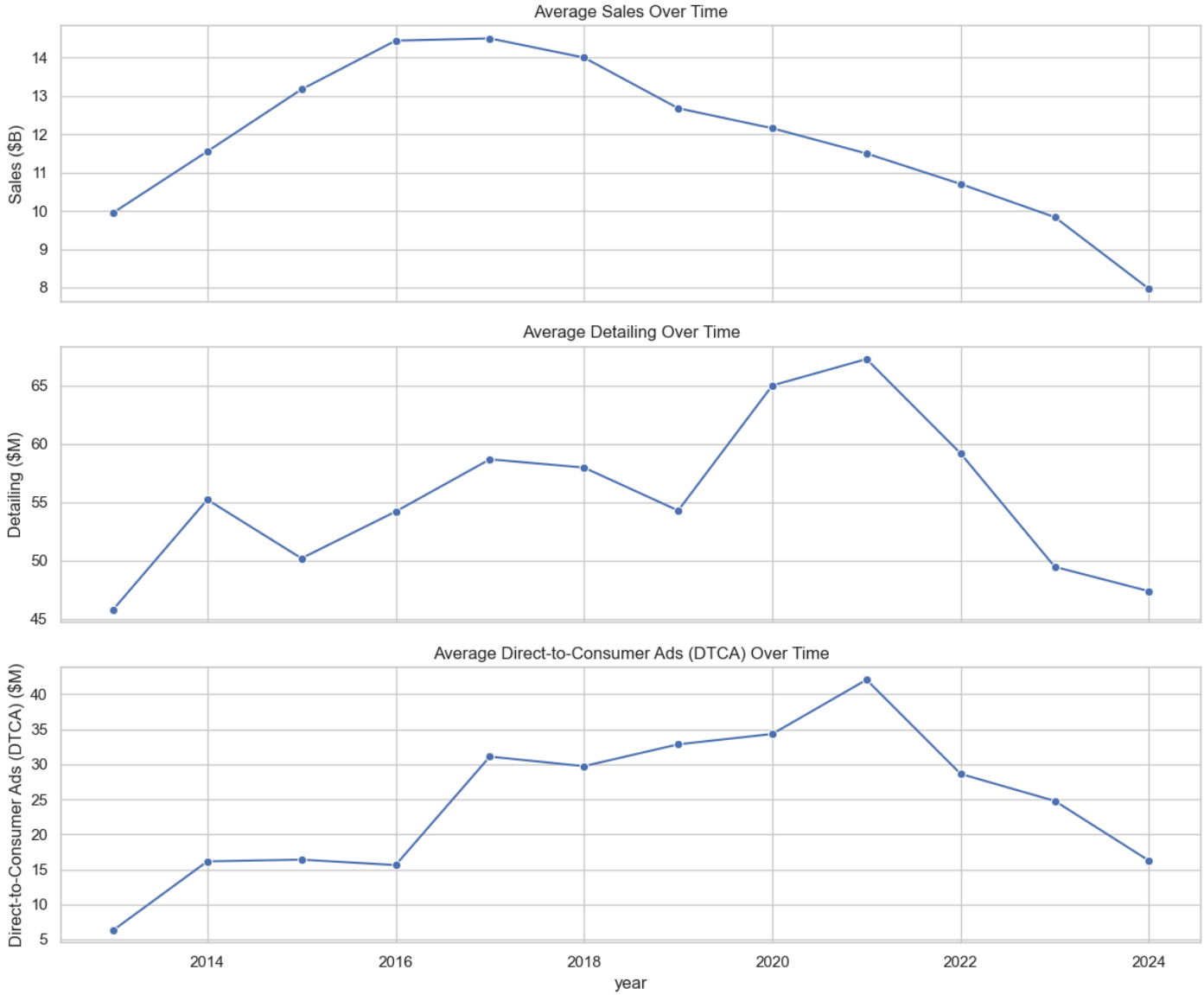
In [246…

```
print("\n[EDA] Analyzing portfolio growth trajectory over time...")

yearly = (
    df.groupby("year", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_ad"]]
    .mean()
)

yearly["sales_yoy_pct"] = yearly["actual_sales"].pct_change() * 100
yearly["detailing_yoy_pct"] = yearly["detailing"].pct_change() * 100
yearly["dtca_yoy_pct"] = yearly["direct_to_consumer_ad"].pct_change() * 100

display(yearly)

# Note: 'yearly' is already aggregated via .mean() above, so these are average values across brands per
fig, axes = plt.subplots(3, 1, figsize=(12, 10), sharex=True)

sns.lineplot(data=yearly, x="year", y="actual_sales", marker="o", ax=axes[0])
axes[0].set_title("Average Sales Over Time")
axes[0].set_ylabel("Sales ($B)")
```

```
sns.lineplot(data=yearly, x="year", y="detailing", marker="o", ax=axes[1])
axes[1].set_title("Average Detailing Over Time")
axes[1].set_ylabel("Detailing ($M)")

sns.lineplot(data=yearly, x="year", y="direct_to_consumer_ad", marker="o", ax=axes[2])
axes[2].set_title("Average Direct-to-Consumer Ads (DTCA) Over Time")
axes[2].set_ylabel("Direct-to-Consumer Ads (DTCA) ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Average sales peaked at ~$14.5B (2017) and declined ~45% to ~$8B by 2024.")
print("Detailing peaked in 2020-2021, Direct-to-Consumer Ads (DTCA) peaked in 2021 before a sharp -34%
print("Marketing spend during this declining phase may be defending existing revenue rather than drivin
print("This distinction matters for how we interpret any efficiency metrics going forward.")
```

```
[EDA] Analyzing portfolio growth trajectory over time...
```

	year	actual_sales	detailing	direct_to_consumer_ad	sales_yoy_pct	detailing_yoy_pct	dtca_yoy_pct
0	2013	9.9486	45.7758	6.3109	NaN	NaN	NaN
1	2014	11.5475	55.2130	16.1484	16.0714	20.6161	155.8812
2	2015	13.1646	50.1752	16.3910	14.0041	-9.1242	1.5025
3	2016	14.4348	54.1879	15.6212	9.6486	7.9974	-4.6969
4	2017	14.4936	58.6714	31.1021	0.4072	8.2740	99.1020
5	2018	13.9928	57.9632	29.7359	-3.4555	-1.2071	-4.3926
6	2019	12.6725	54.2737	32.8522	-9.4353	-6.3652	10.4801
7	2020	12.1514	65.0097	34.3327	-4.1124	19.7812	4.5066
8	2021	11.4936	67.2739	42.0747	-5.4132	3.4829	22.5497
9	2022	10.6978	59.1707	28.6292	-6.9237	-12.0451	-31.9561
10	2023	9.8339	49.4446	24.7726	-8.0753	-16.4374	-13.4710
11	2024	7.9631	47.3541	16.2398	-19.0238	-4.2278	-34.4443

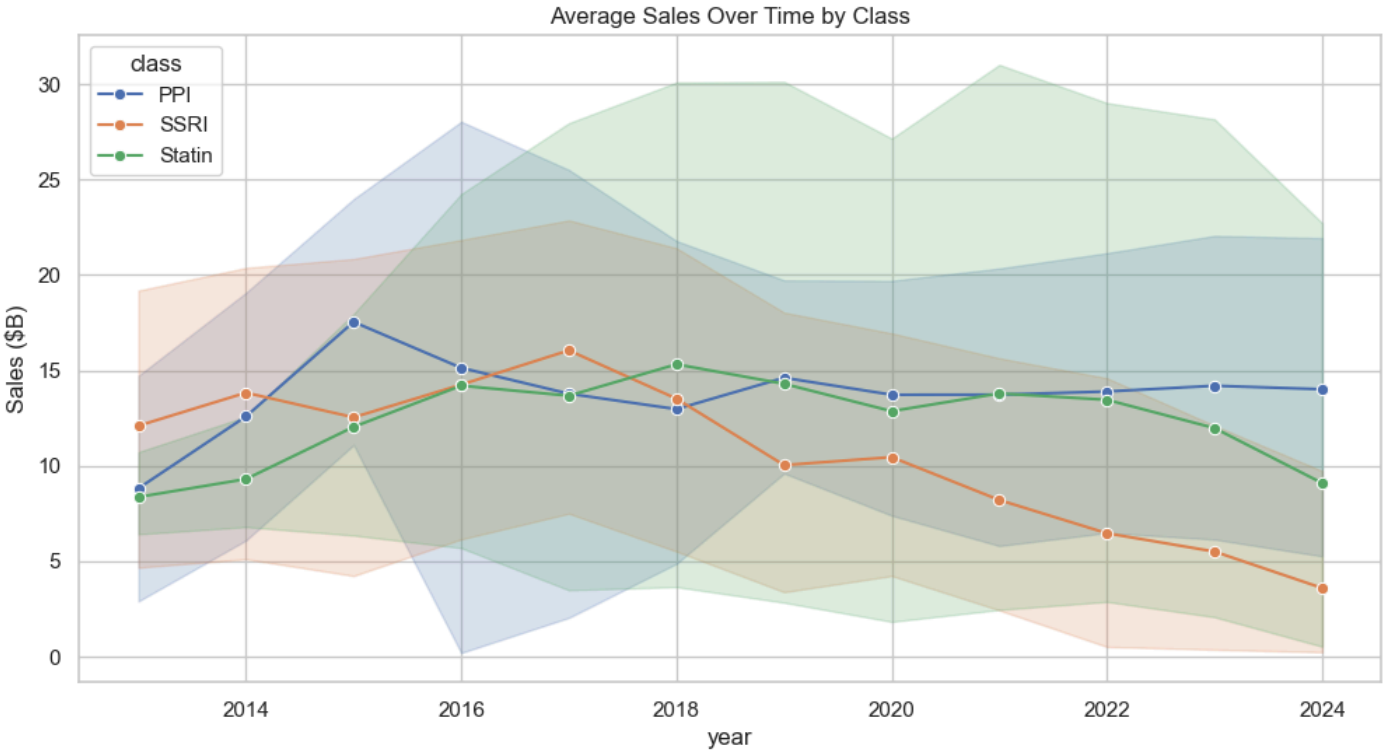



```
[Business Insight]
Average sales peaked at ~$14.5B (2017) and declined ~45% to ~$8B by 2024.
Detailing peaked in 2020-2021, Direct-to-Consumer Ads (DTCA) peaked in 2021 before a sharp -34% drop in
2024.
Marketing spend during this declining phase may be defending existing revenue rather than driving growt
h.
This distinction matters for how we interpret any efficiency metrics going forward.
```

In [247…

```
print("\n[EDA] Comparing class-level sales trends over time...")

plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x="year", y="actual_sales", hue="class", marker="o")
plt.title("Average Sales Over Time by Class")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("PPI (Proton Pump Inhibitors), SSRI (Antidepressants), and Statin (Cholesterol) follow distinct t
print("Statins have the widest range driven by Lipitor; SSRIs peaked ~2018 then declined sharply; PPIs
print("A single portfolio growth narrative is misleading. Strategy reviews should be conducted at the c
```

```
[EDA] Comparing class-level sales trends over time...
```




```
[Business Insight]
PPI (Proton Pump Inhibitors), SSRI (Antidepressants), and Statin (Cholesterol) follow distinct trajector
ies.
Statins have the widest range driven by Lipitor; SSRIs peaked ~2018 then declined sharply; PPIs are most
stable but contracting.
A single portfolio growth narrative is misleading. Strategy reviews should be conducted at the class lev
el.
```

3. Budget Allocation

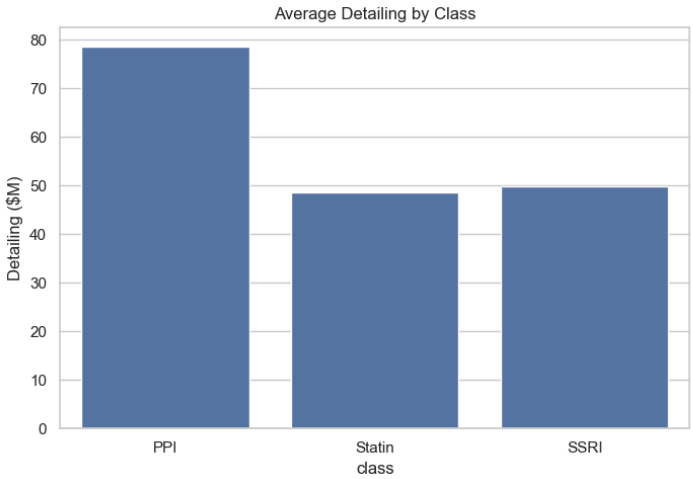
The next question is how the firm allocates detailing and Direct-to-Consumer Ads (DTCA) budgets across classes and brands.

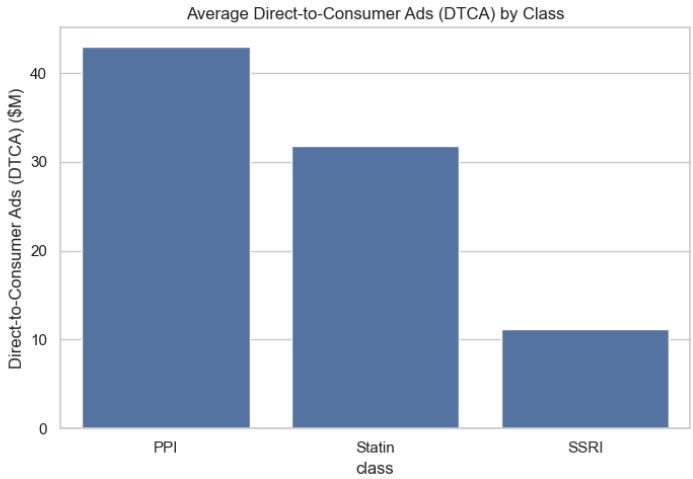
In [248…

```
print("\n[EDA] Summarizing average budget allocation by class...")

class_spend = (
    df.groupby("class", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_ad"]]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(class_spend)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.barplot(data=class_spend, x="class", y="detailing", ax=axes[0])
axes[0].set_title("Average Detailing by Class")
axes[0].set_ylabel("Detailing ($M)")

sns.barplot(data=class_spend, x="class", y="direct_to_consumer_ad", ax=axes[1])
axes[1].set_title("Average Direct-to-Consumer Ads (DTCA) by Class")
axes[1].set_ylabel("Direct-to-Consumer Ads (DTCA) ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("PPI (Proton Pump Inhibitors): highest in both channels ($79M detailing, $43M Direct-to-Consumer
print("Statin (Cholesterol): moderate both ($49M detailing, $32M Direct-to-Consumer Ads (DTCA)).")
print("SSRI (Antidepressants): comparable detailing ($50M) but minimal Direct-to-Consumer Ads (DTCA) ($
print("The firm already differentiates channel strategy by class. Any cross-class reallocation represen
print("a fundamentally different go-to-market approach, not a marginal adjustment.")
```

```
[EDA] Summarizing average budget allocation by class...
```

	class	actual_sales	detailing	direct_to_consumer_ad
0	PPI	13.8355	78.5221	42.9419
2	Statin	12.4964	48.5592	31.7618
1	SSRI	9.5923	49.9067	11.2310






```
[Business Insight]
PPI (Proton Pump Inhibitors): highest in both channels ($79M detailing, $43M Direct-to-Consumer Ads (DTC
A)).
Statin (Cholesterol): moderate both ($49M detailing, $32M Direct-to-Consumer Ads (DTCA)).
SSRI (Antidepressants): comparable detailing ($50M) but minimal Direct-to-Consumer Ads (DTCA) ($11M), he
avily physician-driven.
The firm already differentiates channel strategy by class. Any cross-class reallocation represents
a fundamentally different go-to-market approach, not a marginal adjustment.
```

In [249…

```
print("\n[EDA] Identifying brands with the highest marketing investment...")

brand_spend = (
    df.groupby("agent", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_ad"]]
    .mean()
    .sort_values("detailing", ascending=False)
)

display(brand_spend.head(10))

print("\n[Business Insight]")
print("Agent (brand) strategies diverge sharply:")
print("  Lexapro (SSRI) = $193M detailing, near-zero Direct-to-Consumer Ads (DTCA) (pure physician push
print("  Nexium (PPI) = $153M detailing + $183M Direct-to-Consumer Ads (DTCA) (highest dual-channel inv
print("  Crestor (Statin) = $143M detailing + $119M Direct-to-Consumer Ads (DTCA) (heavy investment des
print("This variation provides natural experiments for disentangling the separate effects of each chann
```

```
[EDA] Identifying brands with the highest marketing investment...
```

	agent	actual_sales	detailing	direct_to_consumer_ad
6	Lexapro	19.1402	192.7008	0.7499
10	Nexium	18.7587	153.0828	182.8566
3	Crestor	7.9550	142.5837	119.2919
7	Lipitor	47.0275	124.0093	80.7884
15	Prevacid	17.8747	92.0651	38.7137
17	Protonix	12.0249	83.2735	3.3244
21	Zoloft	21.1441	80.4162	30.3664
20	Zocor	17.6050	70.3491	54.1879
0	Aciphex	5.8032	66.7499	1.1366
12	Paxil CR	4.8374	53.8639	24.0865

```
[Business Insight]
Agent (brand) strategies diverge sharply:
  Lexapro (SSRI) = $193M detailing, near-zero Direct-to-Consumer Ads (DTCA) (pure physician push);
  Nexium (PPI) = $153M detailing + $183M Direct-to-Consumer Ads (DTCA) (highest dual-channel investor);
  Crestor (Statin) = $143M detailing + $119M Direct-to-Consumer Ads (DTCA) (heavy investment despite mod
erate $8B sales).
This variation provides natural experiments for disentangling the separate effects of each channel.
```

4. Baseline Efficiency

Before formal modeling, it is useful to compute simple descriptive efficiency ratios: how much revenue is associated with each unit of spend?

In [250…

```python
print("\n[EDA] Computing descriptive baseline efficiency by channel...")

efficiency = df.copy()
efficiency["sales_per_detailing"] = np.where(
    efficiency["detailing"] > 0,
    efficiency["actual_sales"] / efficiency["detailing"],
    np.nan
)
efficiency["sales_per_dtca"] = np.where(
    efficiency["direct_to_consumer_ad"] > 0,
    efficiency["actual_sales"] / efficiency["direct_to_consumer_ad"],
    np.nan
)

efficiency_summary = (
    efficiency.groupby("class")[["sales_per_detailing", "sales_per_dtca"]]
    .mean()
    .sort_values("sales_per_detailing", ascending=False)
)

display(efficiency_summary)

avg_eff = efficiency[["sales_per_detailing", "sales_per_dtca"]].mean()
display(avg_eff.to_frame("portfolio_avg_efficiency"))

print("\n[Business Insight]")
print("Detailing shows a more consistent sales-per-dollar relationship across all three classes")
print("(Statin: $0.55B/$M, PPI: $0.39B/$M, SSRI: $0.36B/$M).")
print("Direct-to-Consumer Ads (DTCA) ratios are unreliable. SSRI's apparent $1,979B/$M is an artifact o
print("Directionally, detailing appears to be the more stable revenue driver. Do not use these ratios f
print("budget reallocation; the parametric and ML models will isolate true incremental effects.")
```

```
[EDA] Computing descriptive baseline efficiency by channel...
```

class	sales_per_detailing	sales_per_dtca
Statin	0.5474	59.6006
PPI	0.3913	335.6168
SSRI	0.3591	1,979.3762

	portfolio_avg_efficiency
sales_per_detailing	0.4387
sales_per_dtca	739.4620

```
[Business Insight]
Detailing shows a more consistent sales-per-dollar relationship across all three classes
(Statin: $0.55B/$M, PPI: $0.39B/$M, SSRI: $0.36B/$M).
Direct-to-Consumer Ads (DTCA) ratios are unreliable. SSRI's apparent $1,979B/$M is an artifact of near-z
ero spend denominators.
Directionally, detailing appears to be the more stable revenue driver. Do not use these ratios for
budget reallocation; the parametric and ML models will isolate true incremental effects.
```

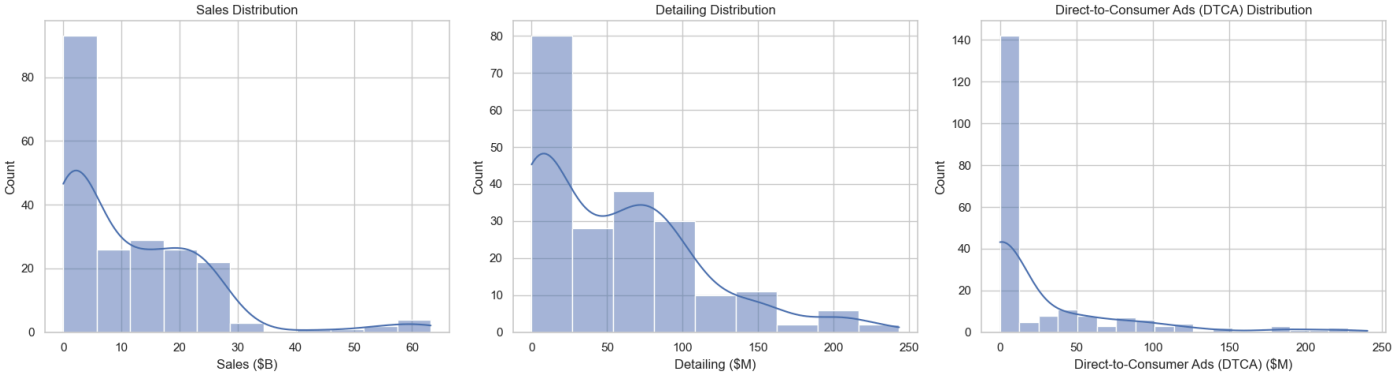
5. Distribution Shape and Market Structure

We now examine whether sales and spend are broadly distributed or concentrated in a small number of observations.

In [251…

```
print("\n[EDA] Visualizing raw distributions of sales and marketing spend...")

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.histplot(df["actual_sales"], kde=True, ax=axes[0])
axes[0].set_title("Sales Distribution")
axes[0].set_xlabel("Sales ($B)")

sns.histplot(df["detailing"], kde=True, ax=axes[1])
axes[1].set_title("Detailing Distribution")
axes[1].set_xlabel("Detailing ($M)")

sns.histplot(df["direct_to_consumer_ad"], kde=True, ax=axes[2])
axes[2].set_title("Direct-to-Consumer Ads (DTCA) Distribution")
axes[2].set_xlabel("Direct-to-Consumer Ads (DTCA) ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Sales, detailing, and Direct-to-Consumer Ads (DTCA) are all heavily skewed to the right.")
print("Most agents (brands) have low-to-moderate values, but a few (like Lipitor and Nexium) have very
print("This concentration means raw averages and simple regressions will be pulled toward these large a
```

```
[EDA] Visualizing raw distributions of sales and marketing spend...
```




```
[Business Insight]
Sales, detailing, and Direct-to-Consumer Ads (DTCA) are all heavily skewed to the right.
Most agents (brands) have low-to-moderate values, but a few (like Lipitor and Nexium) have very high val
ues.
This concentration means raw averages and simple regressions will be pulled toward these large agents (b
rands).
```

In [252…

```
print("\n[EDA] Quantifying skewness in raw variables...")

skewness_raw = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].skew()
display(skewness_raw.to_frame("skewness"))

print("\n[Business Insight]")
print("Skewness values above 1.0 confirm the distributions are significantly lopsided.")
print("Using raw values directly in regression would let a few extreme data points dominate the results
print("This is why we will apply log transformations before modeling.")
```

```
[EDA] Quantifying skewness in raw variables...
```

	skewness
actual_sales	1.7127
detailing	1.0358
direct_to_consumer_ad	2.2988

```
[Business Insight]
Skewness values above 1.0 confirm the distributions are significantly lopsided.
Using raw values directly in regression would let a few extreme data points dominate the results.
This is why we will apply log transformations before modeling.
```

6. Spend Saturation and Diminishing Returns

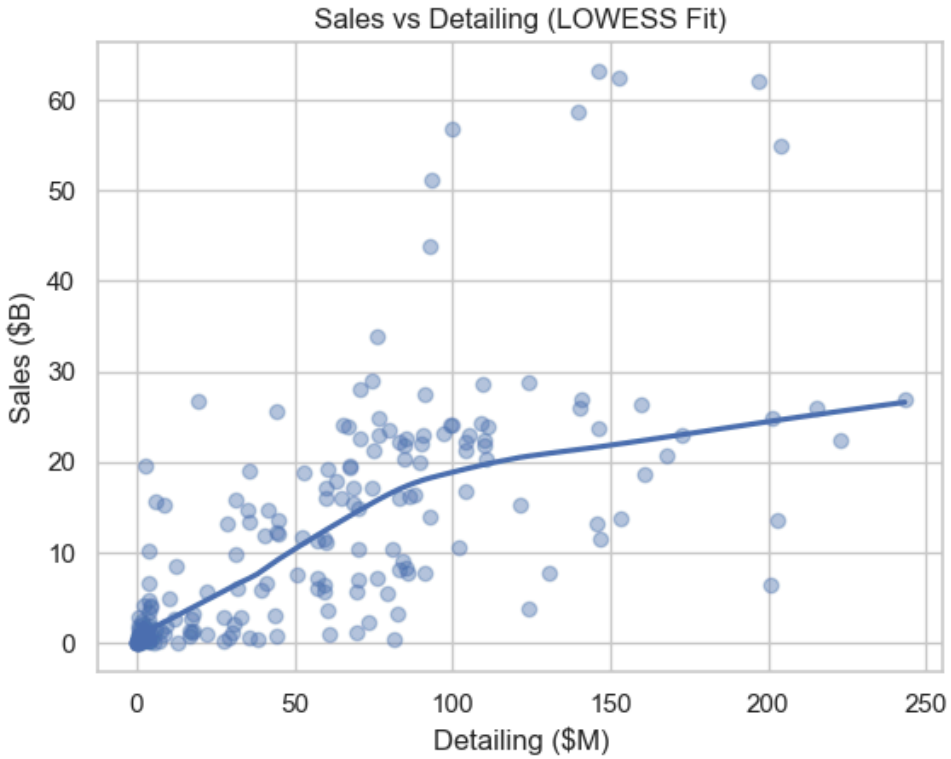
The next business question is whether more spend always produces proportionally more sales.

In [253…

```
print("\n[EDA] Testing for diminishing returns in detailing using LOWESS...")

sns.regplot(
```

```
    data=df,
    x="detailing",
    y="actual_sales",
    lowess=True,
    scatter_kws={"alpha": 0.4}
)

plt.title("Sales vs Detailing (LOWESS Fit)")
plt.xlabel("Detailing ($M)")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("The LOWESS curve shows clear diminishing returns: the first $50M of detailing generates substant
print("more sales lift than the next $50M, and the curve flattens beyond $100M.")
print("For agents (brands) already spending $100M+ (Lexapro, Nexium, Lipitor, Crestor), incremental det
print("dollars yield progressively less return. Reallocation to under-invested agents (brands) could im
```

```
[EDA] Testing for diminishing returns in detailing using LOWESS...
```




```
[Business Insight]
The LOWESS curve shows clear diminishing returns: the first $50M of detailing generates substantially
more sales lift than the next $50M, and the curve flattens beyond $100M.
For agents (brands) already spending $100M+ (Lexapro, Nexium, Lipitor, Crestor), incremental detailing
dollars yield progressively less return. Reallocation to under-invested agents (brands) could improve po
rtfolio ROI.
```

In [254…

```
print("\n[EDA] Approximating where detailing saturation begins...")

df["detailing_bucket"] = pd.qcut(df["detailing"], q=10, duplicates="drop")
bucket_sales = df.groupby("detailing_bucket", observed=False)["actual_sales"].mean().to_frame("avg_sale

display(bucket_sales)

print("\n[Business Insight - Detailing Saturation Threshold]")
print("Bucketed analysis identifies where returns diminish:")
print("  $0-29M: moderate marginal returns (~$1-1.6B added per bucket).")
print("  $29-80M: peak efficiency zone (~$4B per bucket).")
print("  $80-93M: plateau, sales actually dip slightly (saturation onset).")
print("Agents (brands) spending above ~$80M should evaluate whether marginal dollars deliver better ret
```

```
[EDA] Approximating where detailing saturation begins...
```

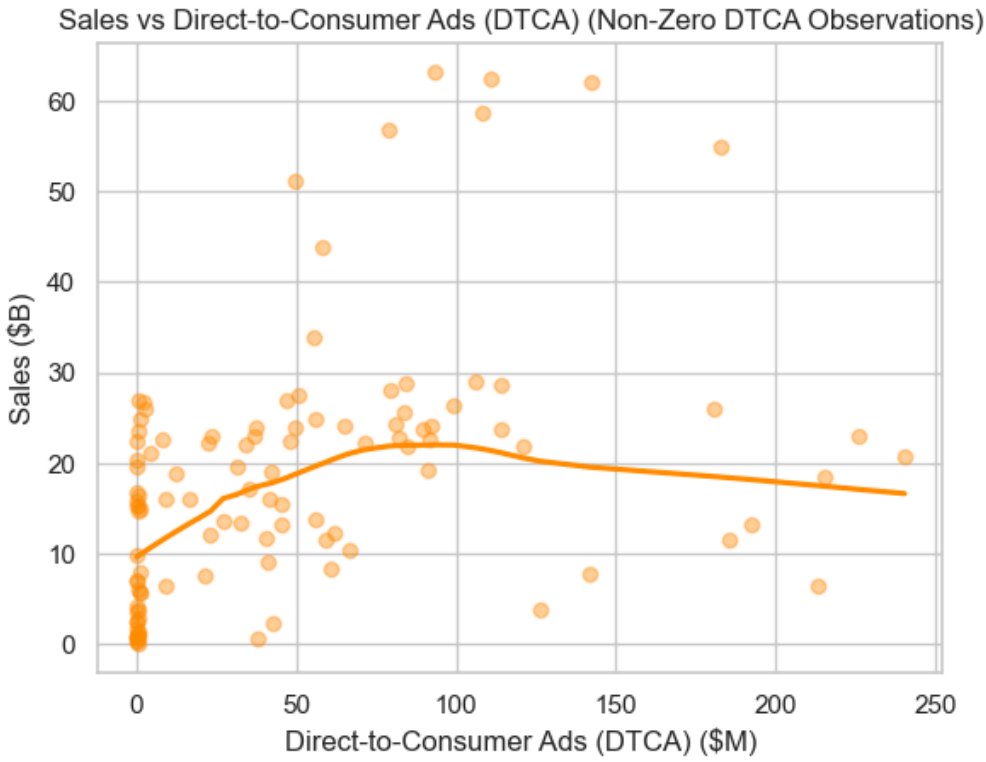
detailing_bucket	avg_sales
(-0.001, 0.639]	0.3719
(0.639, 2.96]	2.0414
(2.96, 6.907]	2.9686
(6.907, 28.631]	4.5371
(28.631, 44.969]	8.5018
(44.969, 67.258]	12.4589
(67.258, 80.046]	16.5847
(80.046, 92.864]	16.3139
(92.864, 134.211]	23.0129
(134.211, 243.33]	29.7583

```
[Business Insight - Detailing Saturation Threshold]
Bucketed analysis identifies where returns diminish:
  $0-29M: moderate marginal returns (~$1-1.6B added per bucket).
  $29-80M: peak efficiency zone (~$4B per bucket).
  $80-93M: plateau, sales actually dip slightly (saturation onset).
Agents (brands) spending above ~$80M should evaluate whether marginal dollars deliver better returns els
ewhere.
```

In [255…

```python
print("\n[EDA] Testing for diminishing returns in Direct-to-Consumer Ads (DTCA) among active advertiser

non_zero_dtca = df[df["direct_to_consumer_ad"] > 0]

sns.regplot(
    data=non_zero_dtca,
    x="direct_to_consumer_ad",
    y="actual_sales",
    lowess=True,
    scatter_kws={"alpha": 0.4},
    color="darkorange"
)

plt.title("Sales vs Direct-to-Consumer Ads (DTCA) (Non-Zero DTCA Observations)")
plt.xlabel("Direct-to-Consumer Ads (DTCA) ($M)")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Among agents (brands) that invest in Direct-to-Consumer Ads (DTCA), higher spend is associated w
print("but the relationship is less consistent than detailing. Similar spend levels produce varying out
print("Agent (brand)-specific factors (market position, class, complementary detailing) likely moderate
print("Direct-to-Consumer Ads (DTCA) effectiveness. The modeling phase will quantify whether the increm
print("is significant after controlling for these factors.")
```

```
[EDA] Testing for diminishing returns in Direct-to-Consumer Ads (DTCA) among active advertisers...
```




```
[Business Insight]
Among agents (brands) that invest in Direct-to-Consumer Ads (DTCA), higher spend is associated with high
er sales,
but the relationship is less consistent than detailing. Similar spend levels produce varying outcomes.
Agent (brand)-specific factors (market position, class, complementary detailing) likely moderate
Direct-to-Consumer Ads (DTCA) effectiveness. The modeling phase will quantify whether the incremental im
pact
is significant after controlling for these factors.
```

7. Class and Brand Differences in Baseline Sales and Response

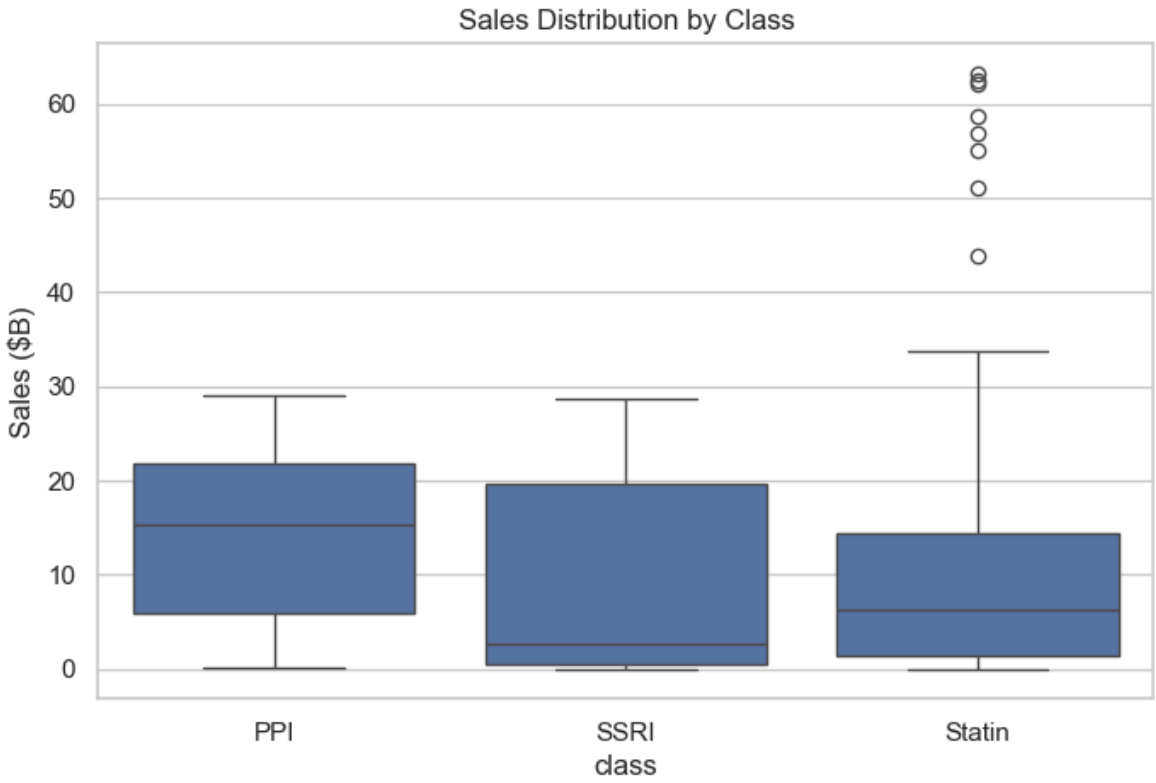
A key strategic issue is whether classes and brands differ structurally in both scale and likely marketing response.

In [256…

```
print("\n[EDA] Comparing baseline sales by class...")

avg_sales_by_class = (
    df.groupby("class", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(avg_sales_by_class)

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="class", y="actual_sales")
plt.title("Sales Distribution by Class")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("PPI (Proton Pump Inhibitors) agents (brands) average $13.8B, Statin (Cholesterol) $12.5B, SSRI (
print("Within-class variation is substantial: Statin agents (brands) range from ~$2B (Mevacor) to ~$47B
print("Marketing budgets calibrated to class averages will over-invest in small agents (brands) and und
```

```
[EDA] Comparing baseline sales by class...
```

	class	actual_sales
0	PPI	13.8355
2	Statin	12.4964
1	SSRI	9.5923



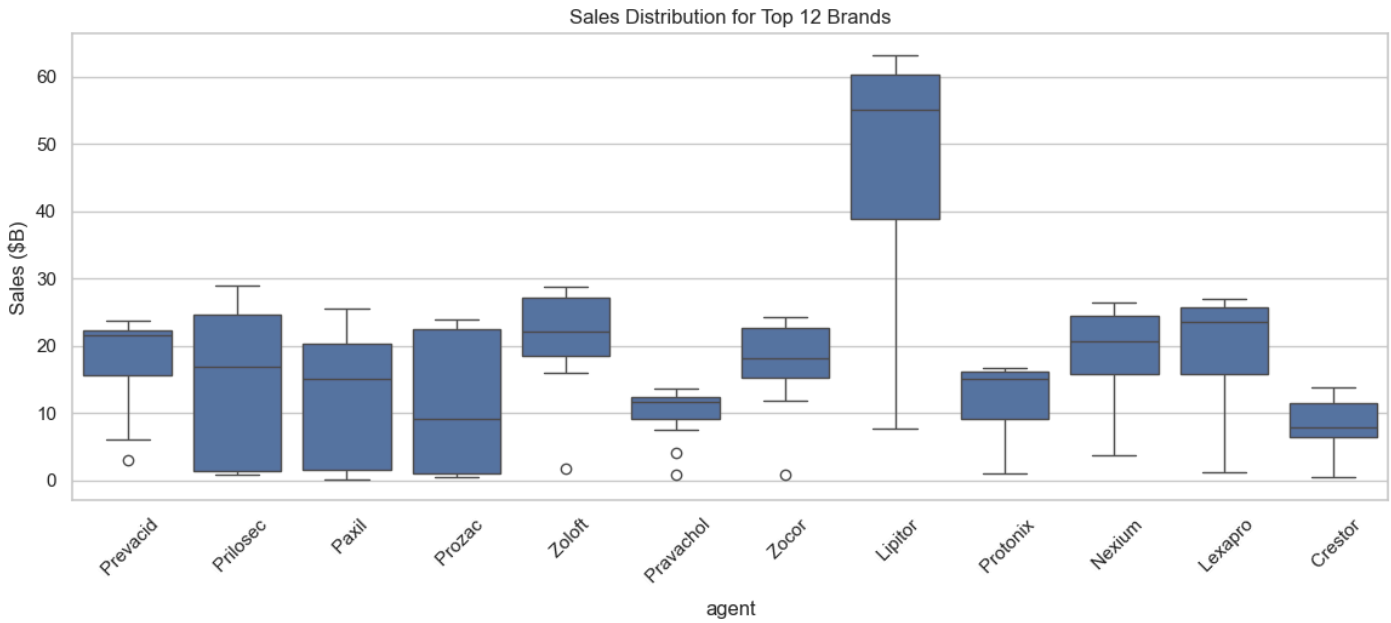

```
[Business Insight]
PPI (Proton Pump Inhibitors) agents (brands) average $13.8B, Statin (Cholesterol) $12.5B, SSRI (Antidepr
essants) $9.6B.
Within-class variation is substantial: Statin agents (brands) range from ~$2B (Mevacor) to ~$47B (Lipito
r).
Marketing budgets calibrated to class averages will over-invest in small agents (brands) and under-inves
t in large ones.
```

In [257…

```
print("\n[EDA] Comparing baseline sales by brand...")

avg_sales_by_agent = (
    df.groupby("agent", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(avg_sales_by_agent.head(15))

top_agents = avg_sales_by_agent.head(12)["agent"].tolist()

plt.figure(figsize=(14, 5))
sns.boxplot(data=df[df["agent"].isin(top_agents)], x="agent", y="actual_sales")
plt.title("Sales Distribution for Top 12 Brands")
plt.ylabel("Sales ($B)")
plt.xticks(rotation=45)
plt.show()

print("\n[Business Insight]")
print("Lipitor ($47B avg) generates ~50x the revenue of Luvox ($0.6B). Even within Statin (Cholesterol)
print("agent (brand) scale varies 10-30x. A large agent (brand) may show high 'efficiency' simply becau
print("not better marketing. The modeling phase will include agent (brand)-level controls to isolate tr
```

```
[EDA] Comparing baseline sales by brand...
```

	agent	actual_sales
7	Lipitor	47.0275
21	Zoloft	21.1441
6	Lexapro	19.1402
10	Nexium	18.7587
15	Prevacid	17.8747
20	Zocor	17.6050
16	Prilosec	14.1557
11	Paxil	12.8184
17	Protonix	12.0249
18	Prozac	11.2791
14	Pravachol	9.9861
3	Crestor	7.9550
2	Celexa	7.8978
0	Aciphex	5.8032
12	Paxil CR	4.8374




```
[Business Insight]
Lipitor ($47B avg) generates ~50x the revenue of Luvox ($0.6B). Even within Statin (Cholesterol),
agent (brand) scale varies 10-30x. A large agent (brand) may show high 'efficiency' simply because of it
s size,
not better marketing. The modeling phase will include agent (brand)-level controls to isolate true marke
ting response.
```

8. Structural Zero-Direct-to-Consumer Ads (DTCA) Regime

A large share of observations show exactly zero Direct-to-Consumer Ads (DTCA), which may reflect a distinct strategic regime rather than just low spend.

In [258…

```
print("\n[EDA] Measuring the prevalence and structure of zero Direct-to-Consumer Ads (DTCA)...")

df["dtca_zero"] = (df["direct_to_consumer_ad"] == 0).astype(int)

zero_dtca = df["dtca_zero"].sum()
total = len(df)

print(f"Zero Direct-to-Consumer Ads (DTCA) observations: {zero_dtca}")
print(f"Total observations: {total}")
print(f"Share of zero-Direct-to-Consumer Ads (DTCA) observations: {zero_dtca / total:.2%}")

print("\nZero Direct-to-Consumer Ads (DTCA) by class:")
display(pd.crosstab(df["class"], df["dtca_zero"]))
```

```
print("\nZero Direct-to-Consumer Ads (DTCA) by year:")
display(pd.crosstab(df["year"], df["dtca_zero"]))

print("\n[Business Insight]")
print("47% of agent (brand)-year observations have zero Direct-to-Consumer Ads (DTCA), reflecting a del
print("SSRI (Antidepressants) agents (brands) are most physician-driven (58% zero Direct-to-Consumer Ad
print("The portfolio operates under two models:")
print("  (1) Detailing-only / physician push, (2) Dual-channel / detailing + Direct-to-Consumer Ads (DT
print("Pooling both groups when evaluating Direct-to-Consumer Ads (DTCA) would be misleading. The model
```

```
[EDA] Measuring the prevalence and structure of zero Direct-to-Consumer Ads (DTCA)...
Zero Direct-to-Consumer Ads (DTCA) observations: 98
Total observations: 207
Share of zero-Direct-to-Consumer Ads (DTCA) observations: 47.34%

Zero Direct-to-Consumer Ads (DTCA) by class:
```

dtca_zero	0	1
class		
PPI	32	16
SSRI	34	47
Statin	43	35

```
Zero Direct-to-Consumer Ads (DTCA) by year:
```

dtca_zero	0	1
year		
2013	3	7
2014	5	6
2015	6	6
2016	6	7
2017	7	8
2018	12	5
2019	12	8
2020	10	11
2021	13	9
2022	13	9
2023	10	12
2024	12	10

```
[Business Insight]
47% of agent (brand)-year observations have zero Direct-to-Consumer Ads (DTCA), reflecting a deliberate
strategic choice, not under-investment.
SSRI (Antidepressants) agents (brands) are most physician-driven (58% zero Direct-to-Consumer Ads (DTC
A)).
The portfolio operates under two models:
  (1) Detailing-only / physician push, (2) Dual-channel / detailing + Direct-to-Consumer Ads (DTCA).
Pooling both groups when evaluating Direct-to-Consumer Ads (DTCA) would be misleading. The modeling phas
e will include a regime indicator.
```

9. Log Transformations for Parametric Modeling

Because the raw variables are highly skewed and the commercial response appears multiplicative, we prepare log variables for the next stage.

In [259…

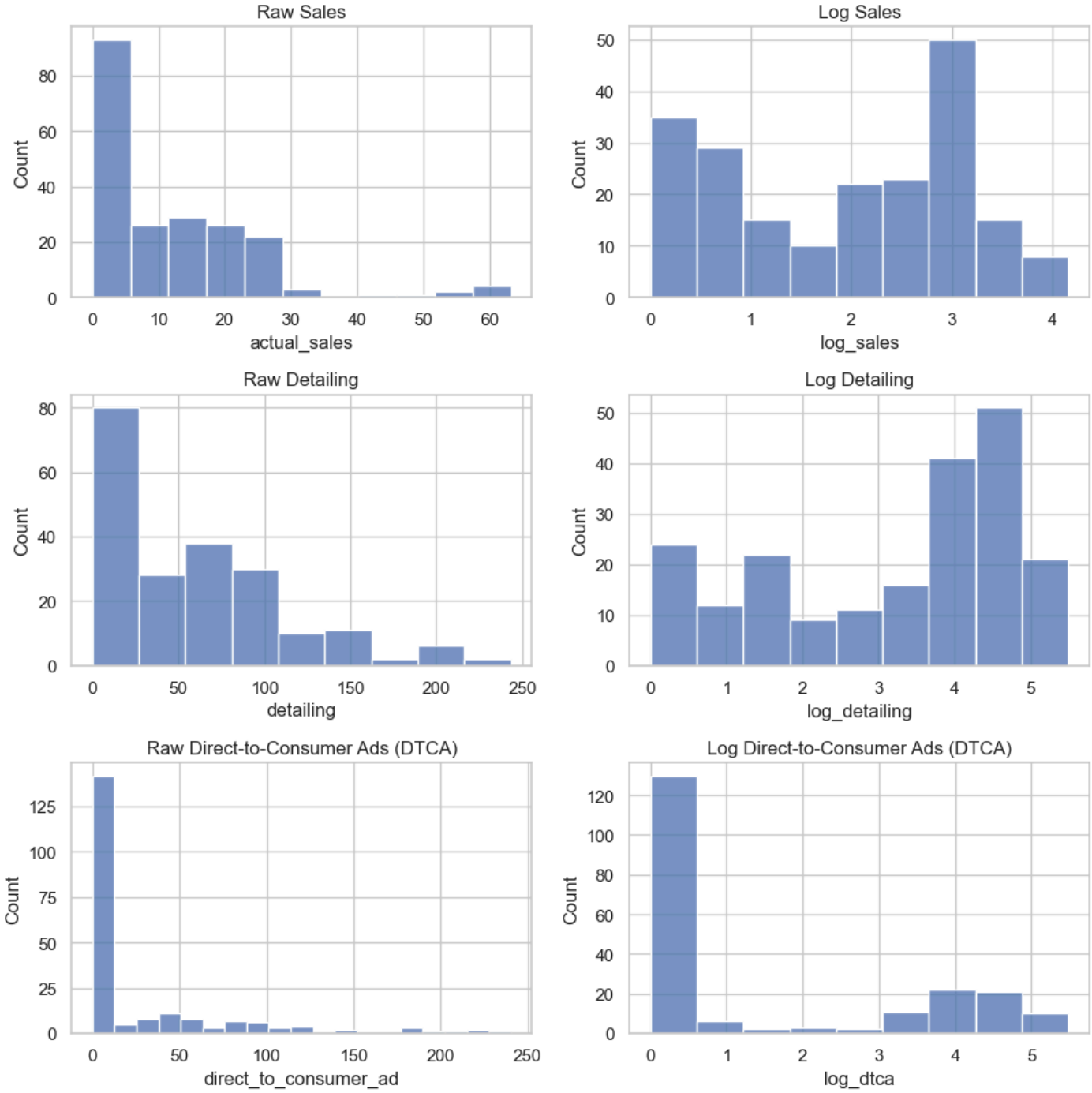
```
print("\n[EDA] Creating log-transformed variables for next-stage modeling...")

df["log_sales"] = np.log1p(df["actual_sales"])
df["log_detailing"] = np.log1p(df["detailing"])
df["log_dtca"] = np.log1p(df["direct_to_consumer_ad"])

print("[EDA] Log variables created: log_sales, log_detailing, log_dtca")
```

```
[EDA] Creating log-transformed variables for next-stage modeling...
[EDA] Log variables created: log_sales, log_detailing, log_dtca
```

In [260…

```python
print("\n[EDA] Comparing raw and log-transformed distributions...")

fig, axes = plt.subplots(3, 2, figsize=(10, 10))

sns.histplot(df["actual_sales"], ax=axes[0, 0])
axes[0, 0].set_title("Raw Sales")

sns.histplot(df["log_sales"], ax=axes[0, 1])
axes[0, 1].set_title("Log Sales")

sns.histplot(df["detailing"], ax=axes[1, 0])
axes[1, 0].set_title("Raw Detailing")

sns.histplot(df["log_detailing"], ax=axes[1, 1])
axes[1, 1].set_title("Log Detailing")

sns.histplot(df["direct_to_consumer_ad"], ax=axes[2, 0])
axes[2, 0].set_title("Raw Direct-to-Consumer Ads (DTCA)")

sns.histplot(df["log_dtca"], ax=axes[2, 1])
axes[2, 1].set_title("Log Direct-to-Consumer Ads (DTCA)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("After log transformation, sales and detailing distributions become much more symmetric and sprea
print("Direct-to-Consumer Ads (DTCA) also improves, but the large spike at zero remains because log(1+0
print("This spike is not a problem. It is a real feature of the data that the model will handle via the
```

```
[EDA] Comparing raw and log-transformed distributions...
```

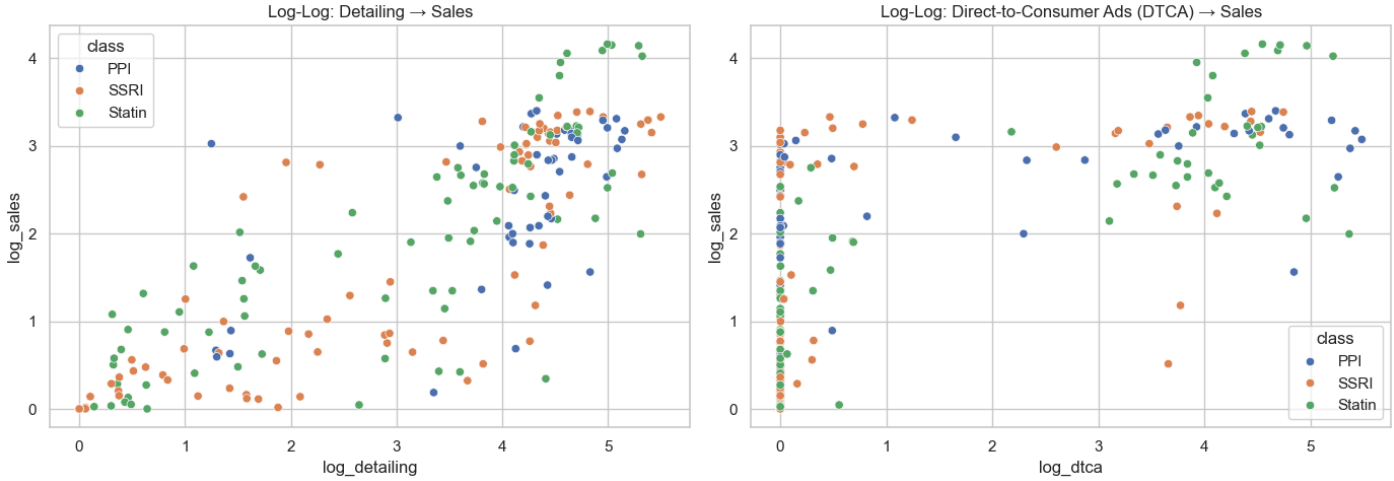



```
[Business Insight]
After log transformation, sales and detailing distributions become much more symmetric and spread out ev
enly.
Direct-to-Consumer Ads (DTCA) also improves, but the large spike at zero remains because log(1+0) = 0 fo
r all zero-Direct-to-Consumer Ads (DTCA) observations.
This spike is not a problem. It is a real feature of the data that the model will handle via the zero-Di
rect-to-Consumer Ads (DTCA) indicator.
```

In [261…

```
print("\n[EDA] Evaluating log-log relationships...")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.scatterplot(data=df, x="log_detailing", y="log_sales", hue="class", ax=axes[0])
axes[0].set_title("Log-Log: Detailing → Sales")

sns.scatterplot(data=df, x="log_dtca", y="log_sales", hue="class", ax=axes[1])
axes[1].set_title("Log-Log: Direct-to-Consumer Ads (DTCA) → Sales")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("In log space, the relationship between marketing spend and sales becomes much closer to a straig
print("This means a log-log regression is appropriate, where coefficients represent elasticities:")
print("e.g., a coefficient of 0.3 means a 1% increase in detailing is associated with a 0.3% increase i
```

```
[EDA] Evaluating log-log relationships...
```

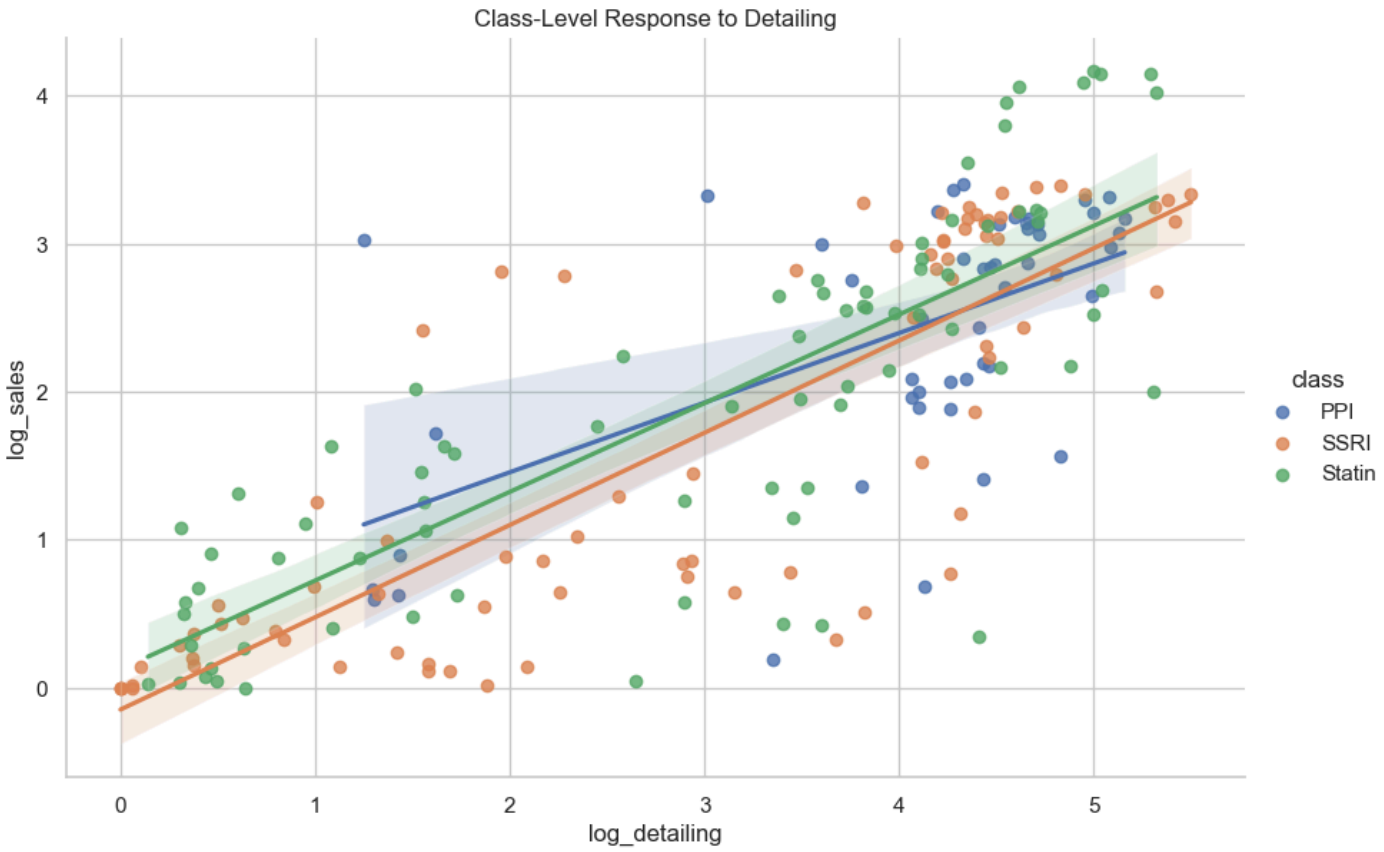



```
[Business Insight]
In log space, the relationship between marketing spend and sales becomes much closer to a straight line.
This means a log-log regression is appropriate, where coefficients represent elasticities:
e.g., a coefficient of 0.3 means a 1% increase in detailing is associated with a 0.3% increase in sales.
```

In [262…

```
print("\n[EDA] Comparing class-level marketing sensitivity in log space...")

sns.lmplot(
    data=df,
    x="log_detailing",
    y="log_sales",
    hue="class",
    height=6,
    aspect=1.5
)

plt.title("Class-Level Response to Detailing")
plt.show()

print("\n[Business Insight]")
print("Statin (Cholesterol) agents (brands) show the steepest detailing response; SSRI (Antidepressants
print("A dollar of detailing generates different returns by class. A uniform budget efficiency target i
print("The modeling phase will quantify class-specific elasticities to support differentiated allocatio
```

```
[EDA] Comparing class-level marketing sensitivity in log space...
```



```
[Business Insight]
Statin (Cholesterol) agents (brands) show the steepest detailing response; SSRI (Antidepressants) the fl
attest.
A dollar of detailing generates different returns by class. A uniform budget efficiency target is inappr
opriate.
The modeling phase will quantify class-specific elasticities to support differentiated allocation strate
gies.
```

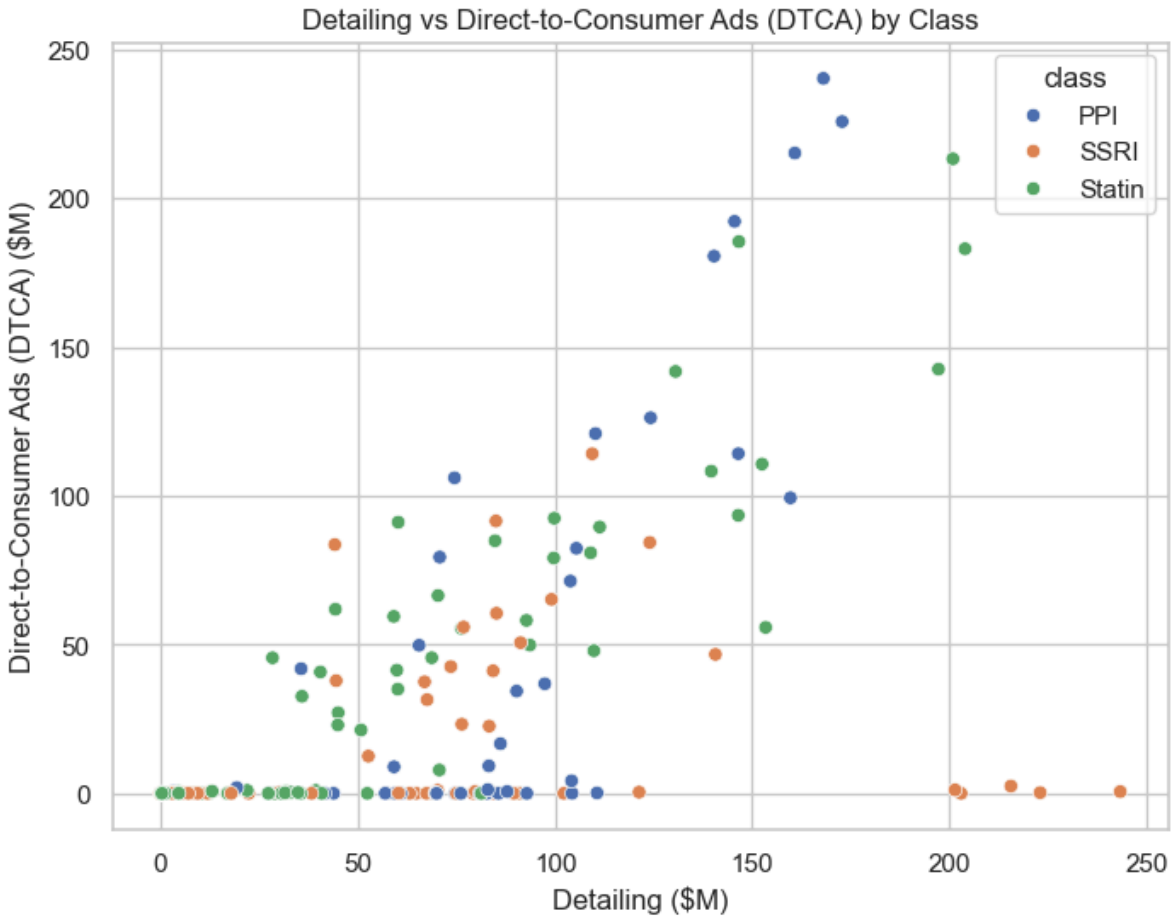
10. Cross-Channel Strategy and Collinearity

Do brands tend to invest in both channels together, or do they specialize?

In [263…

```
print("\n[EDA] Examining the relationship between detailing and Direct-to-Consumer Ads (DTCA)...")

plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=df,
    x="detailing",
    y="direct_to_consumer_ad",
    hue="class"
)
plt.title("Detailing vs Direct-to-Consumer Ads (DTCA) by Class")
plt.xlabel("Detailing ($M)")
plt.ylabel("Direct-to-Consumer Ads (DTCA) ($M)")
plt.show()

print("\n[Business Insight]")
print("PPI (Proton Pump Inhibitors) and Statin (Cholesterol) agents (brands) tend to invest in both cha
print("SSRI (Antidepressants) agents (brands) cluster at high detailing / low Direct-to-Consumer Ads (D
print("Because channels move together for PPI/Statin, isolating each channel's independent effect is ha
```

```
[EDA] Examining the relationship between detailing and Direct-to-Consumer Ads (DTCA)...
```




```
[Business Insight]
PPI (Proton Pump Inhibitors) and Statin (Cholesterol) agents (brands) tend to invest in both channels to
gether.
SSRI (Antidepressants) agents (brands) cluster at high detailing / low Direct-to-Consumer Ads (DTCA).
Because channels move together for PPI/Statin, isolating each channel's independent effect is harder for
those classes.
```

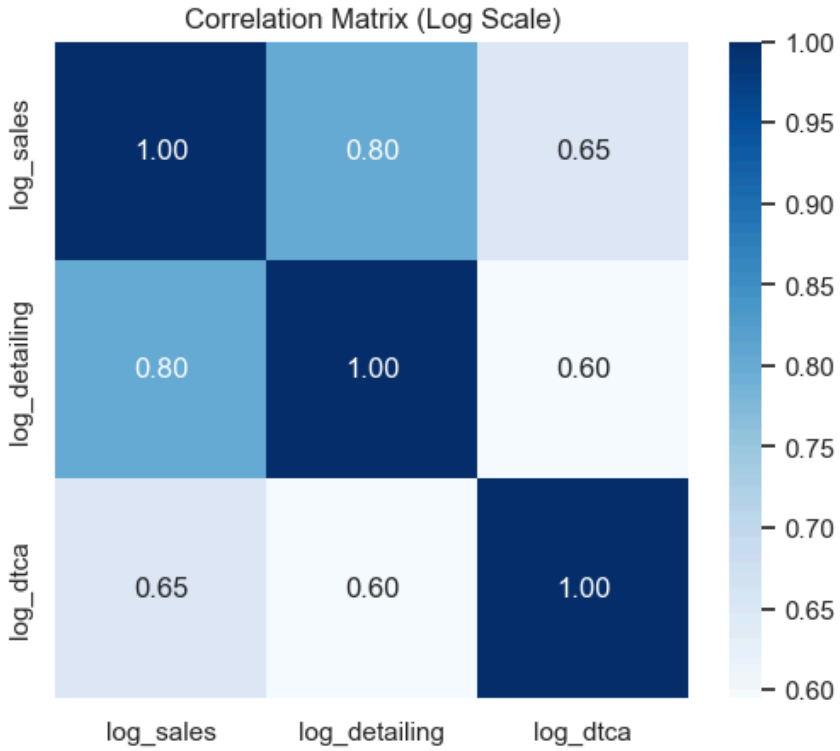
In [264…

```
print("\n[EDA] Calculating correlation matrix for log-transformed variables...")

corr = df[["log_sales", "log_detailing", "log_dtca"]].corr()
```

```
display(corr)

plt.figure(figsize=(6, 5))
sns.heatmap(corr, annot=True, cmap="Blues", fmt=".2f")
plt.title("Correlation Matrix (Log Scale)")
plt.show()

print("\n[Business Insight]")
print("Log-detailing and log Direct-to-Consumer Ads (DTCA) correlate at 0.60 (moderate).")
print("Both correlate positively with sales (detailing: 0.80, Direct-to-Consumer Ads (DTCA): 0.65).")
print("Manageable for modeling, but caution is warranted when attributing sales gains to one channel")
print("when both increased simultaneously for the same agent (brand).")
```

```
[EDA] Calculating correlation matrix for log-transformed variables...
```

	log_sales	log_detailing	log_dtca
log_sales	1.0000	0.8027	0.6482
log_detailing	0.8027	1.0000	0.5953
log_dtca	0.6482	0.5953	1.0000




```
[Business Insight]
Log-detailing and log Direct-to-Consumer Ads (DTCA) correlate at 0.60 (moderate).
Both correlate positively with sales (detailing: 0.80, Direct-to-Consumer Ads (DTCA): 0.65).
Manageable for modeling, but caution is warranted when attributing sales gains to one channel
when both increased simultaneously for the same agent (brand).
```

11. Panel Structure and Within-Brand Dynamics

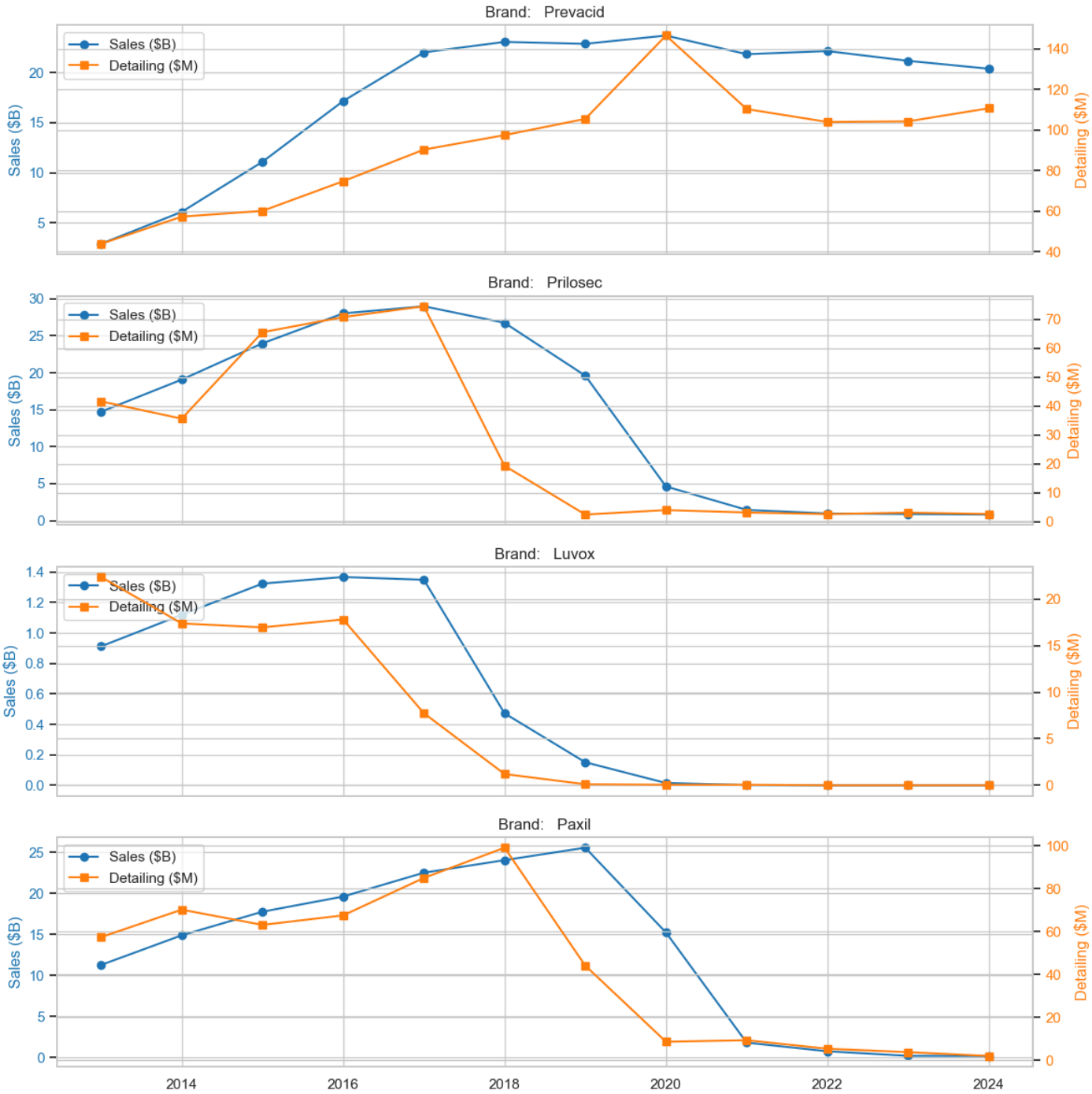
To isolate marketing response credibly, we need to look beyond cross-sectional size and examine within-brand changes over time.

In [265…

```
print("\n[EDA] Exploring within-brand trajectories for selected brands...")

sample_agents = df["agent"].drop_duplicates().head(4).tolist()

fig, axes = plt.subplots(len(sample_agents), 1, figsize=(12, 12), sharex=True)

for i, a in enumerate(sample_agents):
    temp = df[df["agent"] == a].sort_values("year")

    color_sales = "#1f77b4"
    axes[i].plot(temp["year"], temp["actual_sales"], marker="o", color=color_sales, label="Sales ($B)")
    axes[i].set_ylabel("Sales ($B)", color=color_sales)
    axes[i].tick_params(axis="y", labelcolor=color_sales)

    ax_right = axes[i].twinx()
    color_detail = "#ff7f0e"
```

```
    ax_right.plot(temp["year"], temp["detailing"], marker="s", color=color_detail, label="Detailing ($M
    ax_right.set_ylabel("Detailing ($M)", color=color_detail)
    ax_right.tick_params(axis="y", labelcolor=color_detail)

    axes[i].set_title(f"Brand: {a}")
    lines_1, labels_1 = axes[i].get_legend_handles_labels()
    lines_2, labels_2 = ax_right.get_legend_handles_labels()
    axes[i].legend(lines_1 + lines_2, labels_1 + labels_2, loc="upper left")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Individual trajectories confirm that changes in detailing over time correspond with sales change
print("Prevacid ramped detailing 2013-2017, sales followed; Paxil spiked 2018-2019, then both declined;
print("Luvox dropped detailing to near zero after 2018, sales declined in parallel.")
print("This within-agent (brand) evidence suggests marketing spend has a real association with sales")
print("beyond the effect of agent (brand) scale alone.")
```

```
[EDA] Exploring within-brand trajectories for selected brands...
```




```
[Business Insight]
Individual trajectories confirm that changes in detailing over time correspond with sales changes:
Prevacid ramped detailing 2013-2017, sales followed; Paxil spiked 2018-2019, then both declined;
Luvox dropped detailing to near zero after 2018, sales declined in parallel.
This within-agent (brand) evidence suggests marketing spend has a real association with sales
beyond the effect of agent (brand) scale alone.
```

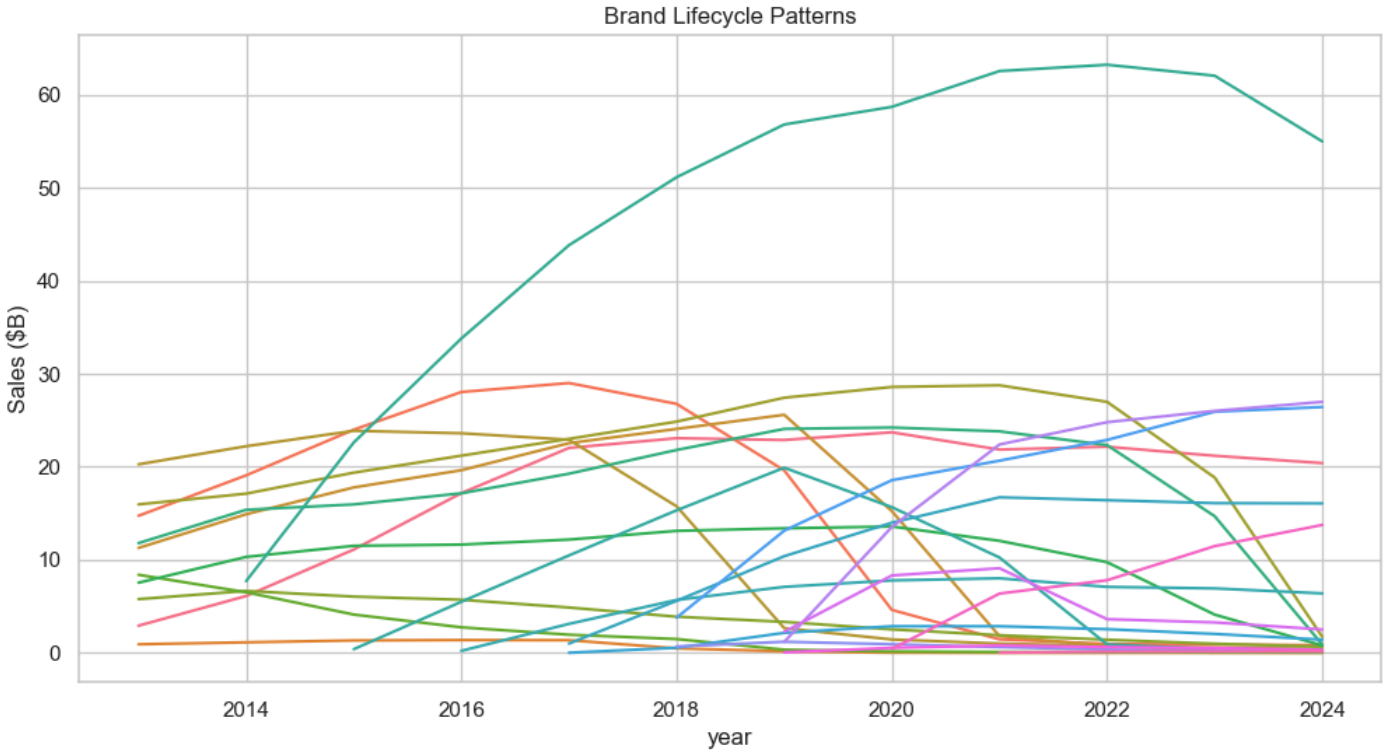
In [266…

```
print("\n[EDA] Inferring brand lifecycle patterns...")
```

```python
agent_lifecycle = (
    df.groupby("agent")
    .agg(
        start_year=("year", "min"),
        end_year=("year", "max"),
        avg_sales=("actual_sales", "mean")
    )
    .sort_values("start_year")
)

display(agent_lifecycle.head(15))

plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x="year", y="actual_sales", hue="agent", legend=False)
plt.title("Brand Lifecycle Patterns")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Agents (brands) are at different lifecycle stages: Lipitor plateaued, Luvox is in steady decline
print("newer entrants like Crestor are still ramping. A growing agent (brand) may appear to have high m
print("simply because of organic growth, and a declining one may appear inefficient despite sound strat
print("The modeling phase will include year and agent (brand) controls to separate lifecycle trends fro
```

```
[EDA] Inferring brand lifecycle patterns...
```

agent	start_year	end_year	avg_sales
Zoloft	2013	2024	21.1441
Prozac	2013	2024	11.2791
Prilosec	2013	2024	14.1557
Lescol	2013	2024	3.6356
Prevacid	2013	2024	17.8747
Pravachol	2013	2024	9.9861
Luvox	2013	2024	0.5595
Mevacor	2013	2024	2.1499
Zocor	2013	2024	17.6050
Paxil	2013	2024	12.8184
Lipitor	2014	2024	47.0275
Celexa	2015	2024	7.8978
Aciphex	2016	2024	5.8032
Lescol XL	2017	2024	1.7925
Protonix	2017	2024	12.0249




```
[Business Insight]
Agents (brands) are at different lifecycle stages: Lipitor plateaued, Luvox is in steady decline,
newer entrants like Crestor are still ramping. A growing agent (brand) may appear to have high marketing
ROI
simply because of organic growth, and a declining one may appear inefficient despite sound strategy.
The modeling phase will include year and agent (brand) controls to separate lifecycle trends from true m
arketing impact.
```

12. Outliers and High-Leverage Observations

Before modeling, we identify whether a few brands or campaigns are likely to dominate results.

In [267…

```
print("\n[EDA] Reviewing highest-leverage observations...")

print("Top 5 by sales:")
display(df.nlargest(5, "actual_sales")[["class", "agent", "year", "actual_sales", "detailing", "direct_

print("Top 5 by detailing:")
display(df.nlargest(5, "detailing")[["class", "agent", "year", "actual_sales", "detailing", "direct_to_

print("Top 5 by Direct-to-Consumer Ads (DTCA):")
display(df.nlargest(5, "direct_to_consumer_ad")[["class", "agent", "year", "actual_sales", "detailing",

print("\n[Business Insight]")
print("Three agents (brands) dominate the extremes: Lipitor (top sales $55-63B),")
print("Lexapro (top detailing $203-243M), Nexium (top Direct-to-Consumer Ads (DTCA) $192-240M).")
print("These are not anomalies; they reflect real market positions.")
print("Any portfolio-level finding that changes when one of these agents (brands) is removed is really"
print("a finding about that agent (brand), not the portfolio. Robustness checks will flag this distinct
```

```
[EDA] Reviewing highest-leverage observations...
Top 5 by sales:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
159	Statin	Lipitor	2022	63.2186	146.4874	93.4800
137	Statin	Lipitor	2021	62.5406	152.5107	110.7082
181	Statin	Lipitor	2023	62.0389	197.2503	142.6181
115	Statin	Lipitor	2020	58.6918	139.6403	108.3209
94	Statin	Lipitor	2019	56.7952	99.7099	79.1343

```
Top 5 by detailing:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
191	SSRI	Lexapro	2024	26.9766	243.3303	0.5963
125	SSRI	Lexapro	2021	22.3924	223.0231	0.2595
169	SSRI	Lexapro	2023	26.0019	215.6069	2.4667
203	Statin	Lipitor	2024	54.9727	203.9765	183.0804
104	SSRI	Lexapro	2020	13.5176	202.9746	0.0000

```
Top 5 by Direct-to-Consumer Ads (DTCA):
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
120	PPI	Nexium	2021	20.6499	168.0683	240.3550
142	PPI	Nexium	2022	22.8833	172.8186	225.8439
99	PPI	Nexium	2020	18.5523	160.8279	215.2616
134	Statin	Crestor	2021	6.3594	200.9031	213.3158
79	PPI	Nexium	2019	13.1240	145.5416	192.2922

```
[Business Insight]
Three agents (brands) dominate the extremes: Lipitor (top sales $55-63B),
Lexapro (top detailing $203-243M), Nexium (top Direct-to-Consumer Ads (DTCA) $192-240M).
These are not anomalies; they reflect real market positions.
Any portfolio-level finding that changes when one of these agents (brands) is removed is really
a finding about that agent (brand), not the portfolio. Robustness checks will flag this distinction.
```

13. EDA Takeaways

In [268…

```python
print("\n" + "="*90)
print("EDA EXECUTIVE SUMMARY")
print("="*90)
print("""
Q1 - REVENUE CONCENTRATION:
  Top 5 agents (brands) account for ~52% of revenue. Lipitor alone = ~20%.
  Budget decisions around these high-revenue agents (brands) carry outsized portfolio risk.

Q2 - GROWTH TRAJECTORY:
  Portfolio sales peaked ~2017 ($14.5B avg), declined ~45% to ~$8B by 2024.
  PPI (Proton Pump Inhibitors), SSRI (Antidepressants), Statin (Cholesterol) are all contracting
  at different rates. Marketing ROI must be evaluated against a shrinking baseline.

Q3 - BUDGET ALLOCATION:
  Channel strategy varies by class: PPI = dual-channel, SSRI = detailing-only, Statin = moderate both.
  At the agent (brand) level, Lexapro, Nexium, and Crestor dominate total spend.

Q4 - CHANNEL EFFICIENCY:
  Detailing shows a more consistent sales relationship across classes.
  Direct-to-Consumer Ads (DTCA) ratios are unreliable due to 47% structural zeros.
  Formal modeling is required before any reallocation decisions.

Q5 - DIMINISHING RETURNS:
  Detailing saturates around $80-93M. Incremental dollars above this yield progressively less.
  Direct-to-Consumer Ads (DTCA) is positive but inconsistent; agent (brand)-specific factors
  likely moderate effectiveness.

Q6 - CLASS & AGENT (BRAND) STRUCTURE:
  Statin agents (brands) show steepest detailing response; SSRIs the flattest.
  Agent (brand) scale varies 50x within the portfolio. Uniform efficiency targets are inappropriate.

Q7 - MARKETING REGIMES:
  47% of observations use zero Direct-to-Consumer Ads (DTCA), reflecting a deliberate
  physician-only strategy, not under-investment. SSRI agents (brands) are most physician-driven
  (58% zero Direct-to-Consumer Ads (DTCA)).

Q8 - PANEL STRUCTURE:
  Unbalanced panel: 5 agents (brands) entered after 2016, Pexeva has only 4 years.
  Within-agent (brand) trajectories confirm spend-sales co-movement, supporting causal modeling.
""")
```

```
==========================================================================================
EDA EXECUTIVE SUMMARY
==========================================================================================

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```

Next Steps for Parametric Modeling

This EDA has already prepared the key features needed for the next stage:

- `log_sales`
- `log_detailing`
- `log_dtca`
- `dtca_zero`

Recommended first parametric baseline

[\log(1 + Sales) \sim \log(1 + Detailing) + \log(1 + DTCA) + Class + Agent + Year]

Machine Learning Models:

[Gradient boosting / random forest / regularized regression for nonlinearities and interactions.]

Robustness & Validation:

Time-based split (Train 2013-2021, Val 2022-2023, Test 2024). Exclusion tests for Lipitor, Lexapro, Nexium. Parametric vs. ML comparison.

Extensions suggested by the EDA

- class × channel interaction terms
- `dtca_zero` as a binary regime indicator
- nonlinear response checks if residual diagnostics suggest misfit
- robustness checks for high-leverage brands

In [269…

```python
print("\n[EDA] Saving processed dataset for the next modeling stage...")

df.to_csv("data/processed_data.csv", index=False)

print("[EDA] Saved processed file: data/processed_data.csv")
```

```
[EDA] Saving processed dataset for the next modeling stage...
[EDA] Saved processed file: data/processed_data.csv
```

In [269…